

ComicCraft - AI Comic Story Creator using Gemini Models

4. Requirement Analysis

Technology Track: Generative AI with Google

1. Purpose

This section defines what ComicCraft must do (functional requirements) and how well it must perform (non-functional requirements), along with the technical setup needed to build it.

2. Functional Requirements

ID	Requirement	Priority
FR-1	Accept a story idea typed by the user as plain text.	Must-have
FR-2	Break the story into an ordered set of panel scenes using Gemini text generation.	Must-have
FR-3	Generate a short caption or dialogue line for every panel.	Must-have
FR-4	Generate matching artwork per panel using Gemini image generation, with a consistent style.	Must-have
FR-5	Let the user choose an art style before generating.	Should-have
FR-6	Show a preview of the full comic strip before final export.	Must-have
FR-7	Allow the finished comic to be downloaded as an image file.	Should-have
FR-8	Allow regenerating a single panel without affecting the rest of the comic.	Should-have
FR-9	Optionally keep a history of previously generated comics.	Nice-to-have

3. Non-Functional Requirements

ID	Requirement	Category
NFR-1	A 4-6 panel comic should be generated within about a minute.	Performance
NFR-2	The interface must work well on both desktop and mobile browsers.	Usability
NFR-3	Generated text and images must pass content-safety checks.	Safety
NFR-4	API keys must be kept server-side, never exposed to the browser.	Security

ID	Requirement	Category
NFR-5	The app must show a clear error message, not crash, if the AI service fails.	Reliability

4. Technical Requirements

4.1 Software

- Frontend: React.js, HTML5, CSS3
- Backend: Node.js with Express.js
- AI Services: Google Gemini API — text generation and image generation
- Optional Storage: Firebase Firestore for saving comic history

4.2 Hardware

- A standard development machine with a stable internet connection
- No special hardware required on the end-user side, since it runs in a browser

5. Key Use Cases

Use Case	Description
Generate a Comic	User submits a story idea and receives a finished illustrated comic strip.
Select Art Style	User picks the visual style applied consistently across all panels.
Redo a Panel	User requests a new version of a single panel only.
Export Comic	User downloads the finished comic to their device.

6. Assumptions and Constraints

- The system depends on Gemini API availability and usage quota.
- An active internet connection is required for every generation request.
- Output quality depends on the Gemini image model's capability at build time.

7. Summary

These requirements set the functional and technical baseline that guided the design and development of ComicCraft, keeping the focus on fast, safe, and visually consistent comic generation.